

MALAWI UNIVERSITY OF SCIENCE AND TECHNOLOGY

MALAWI INSTITUTE OF TECHNOLOGY

APPLIED STUDIES DEPARTMENT

2019/20 END OF FIRST SEMESTER SUPPLEMENTARY EXAMINATIONS

PHYS-111: MECHANICS AND THERMAL PROPERTIES OF MATTER

DATE: 22nd February, 2021 TIME: OPEN BOOK (9:00 am-5:00 pm)

Examiners: F. Benard, R. Chimombo, C. Daka, S. Chisale, P. Mnjeza, C. Malanda.

INSTRUCTIONS

1. Write your **Examination number** on each page of your answer sheets.
 2. This paper contains **7 pages** and **EIGHT (8) QUESTIONS**, please check
 3. Answer **ALL** questions.
 4. Marks allocated for each question are indicated against it.
 5. Start each question on a fresh page
 6. Round off your final answer to 2 decimal places.
 7. Show your work clearly.
 8. Take pictures of your work (**Hand written solutions**) and upload them on Moodle when done.
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CONSTANT(S)

Acceleration due to gravity	g	9.8 m/s^2
Boltzmann's constant	k_B	$1.38 \times 10^{-23} \text{ J/K}$
Avogadro's number	N_A	$6.02 \times 10^{23} / \text{mol}$
Universal gas constant	R	8.318 J/mol.K
Gravitation constant	G	$6.67 \times 10^{-11} \text{ N.m}^2/\text{kg}^2$
Specific heat of water	c_w	$4186 \text{ J/kg.}^\circ\text{C}$
Specific heat of ice	c_i	$2090 \text{ J/kg.}^\circ\text{C}$
Specific heat of steam	c_s	$2010 \text{ J/kg.}^\circ\text{C}$
Latent heat of fusion of water	L_f	$3.33 \times 10^5 \text{ J/kg}$
Latent heat of vaporization of water	L_v	$2.26 \times 10^6 \text{ J/kg}$
Radius of the Earth	R_E	$6.37 \times 10^6 \text{ m}$
Mass of the Earth	M_E	$5.98 \times 10^{24} \text{ kg}$

Question 1 [10 Marks]

- a. Define basic physical quantities [2]
- b. Find the order of magnitude of 0.0029 [3]
- c. The speed of sound (u) in a gas might plausibly depends on pressure p , the density ρ , and the volume (V) of the gas. Use dimensional analysis to derive the formula for the speed of sound (u). [5]

Question 2 [12 Marks]

- a. A car starts from rest and accelerates uniformly over a time of 5.21 seconds for a distance of 110 m. determine the acceleration of the car. [3]
- b. An arrow is shot straight up in the air at an initial speed of 15 m/s. after how much time is the arrow heading downward at a speed of 8 m/s [3]
- c. A ball is thrown vertically upward with a speed of 25.0 m/s.
 - i. How high does it rise? [3]
 - ii. How long does it take to reach its highest point? [3]

Question 3 [12 Marks]

- a. One car travels due east at 40 km/h and a second car travels north at 40 km/h. Are their velocities equal? Explain. [2]
- b. A firefighter 50 m away from a burning building directs a stream of water from a fire hose at an angle of 30° above the horizontal. If the speed of the stream is 40 m/s, at what height will the water strike the building? [5]
- c. A ball is thrown straight upward and returns to the thrower's hand after 3.00 s in the air. A second ball is thrown at an angle of 30.0°

with the horizontal. At what speed must the second ball be thrown so that it reaches the same height as the one thrown vertically. [5]

Question 4 [10 Marks]

- a. Explain why newton's first law of motion is also known as the law of inertia. [2]
- b. A block of mass 5 kg on a rough, horizontal surface is connected to a ball of mass 10 kg by a lightweight cord over a lightweight, frictionless pulley, as shown in figure 1. A force of magnitude 50 N at an angle 30° with the horizontal is applied to the block as shown. The coefficient of kinetic friction between the block and the surface is 0.12 . Determine the magnitude of the acceleration of the two objects.

[8]

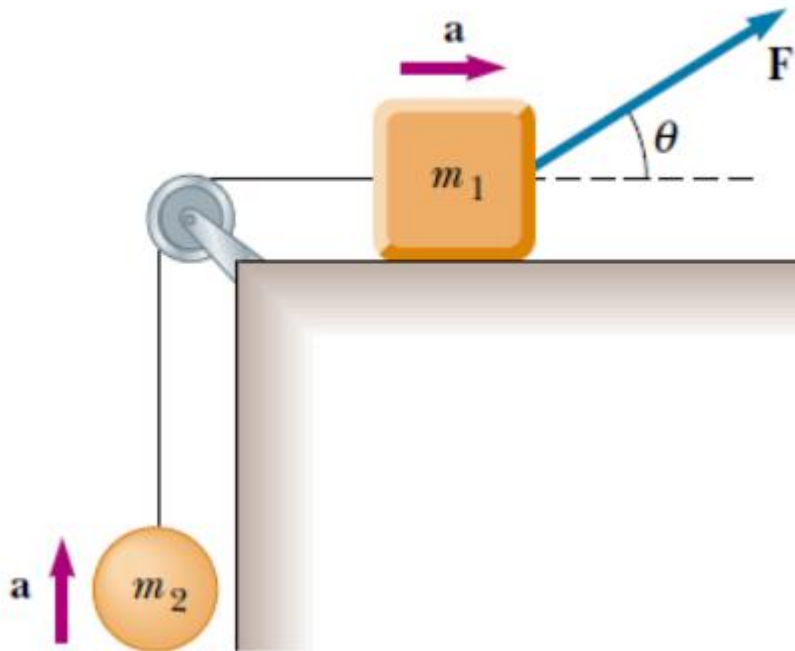


Figure 1

Question 5 [16 Marks]

- a. State work-kinetic energy theorem [2]
- b. A 5.0 kg object is moving with a speed of 3.0 m/s. Find the kinetic energy of the object. [3]
- c. A child and a sled with a combined mass of 50.0 kg slide down a frictionless slope. If the sled starts from rest and has a speed of 3.00 m/s at the bottom, what is the height of the hill? [4]
- d. A 65 kg trampoline artist jumps vertically upward from the top of a platform with a speed of 5.0 m/s, figure 2.
 - i. How fast is he going as he lands on the trampoline, 3.0 m below? [3]
 - ii. If the trampoline behaves like a spring with spring stiffness constant 6.2×10^4 N/m, how far does he depress it? [4]

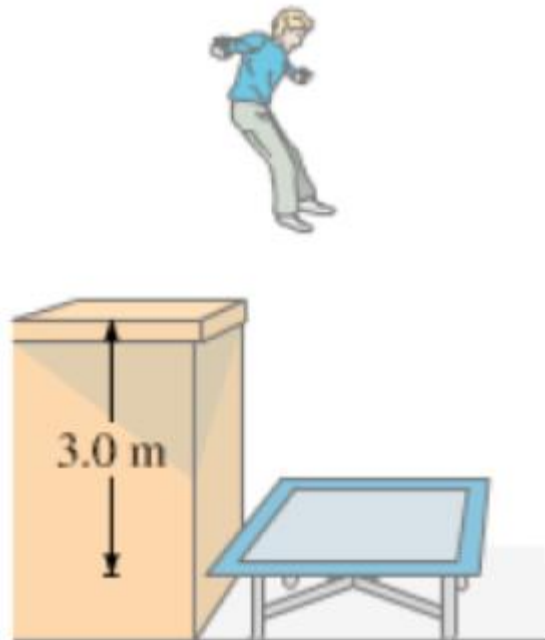


Figure 2

Question 6 [14 Marks]

- a. State impulse-momentum theorem. [2]
- b. A tennis ball of mass 0.06 kg and speed 25 m/s strikes a wall at 45° angle and rebounds with the same speed at 45° , figure 3. What is the impulse given to the ball? [3]

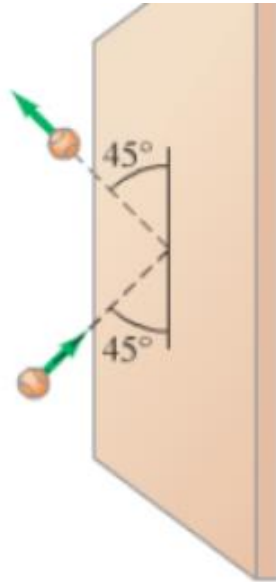


Figure 3

- c. An eagle of mass 4.3 kg moving with speed 7.8 m/s is on a collision course with a second eagle of mass 5.6 kg moving at 10.2 m/s in a direction perpendicular to the direction of the first eagle. After they collide, they hold onto one another.
- i. Find their speed after collision [6]
- ii. Find their direction after the collision [3]

Question 7 [15 Marks]

- a. A racing car travels on a circular track of radius 250 m. If the car moves with a constant linear speed of 45.0 m/s, find its angular speed [3]
- b. A wheel starts from rest and rotates with constant angular acceleration to reach an angular speed of 12.0 rad/s in 3.00 s. Find
- i. the magnitude of the angular acceleration of the wheel. [3]
 - ii. the angle in radians through which it rotates in this time. [3]
- c. A ball on the end of a string is revolved at a uniform rate in a vertical circle of radius 72 cm. If its speed is 4 m/s and its mass is 0.3 kg, calculate the tension in the string when the ball is:
- i. At the top of its path [3]
 - ii. At the bottom of its path [3]

Question 8 [11 Marks]

- a. State zeroth law of thermodynamics [2]
- b. Convert the following temperatures to degree Fahrenheit
- i. 55 K [2]
 - ii. 100 K [2]
- c. An extraterrestrial scientist invents a temperature scale such that water freezes at -25°E and boils at 275°E , where E stands for an extraterrestrial scale. Find an equation that relates temperature in $^{\circ}\text{E}$ to temperature in $^{\circ}\text{C}$. [5]

END OF QUESTIONS